

The effects of temperature stress and desiccation on two intertidal crabs, *Hemigrapsus oregonensis* and *Carcinus maenus*.

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Abstract

With the rising concern over the potential for European green crab (*Carcinus maenus*), understanding how their physiology compares to native species, such as the hairy shore crab (*Hemigrapsus oregonensis*), especially as environmental conditions are altered by climate change, has become a vital topic. Factors such as temperature can change the metabolic rate of organisms, especially ectotherms, to a point that it can become stressful. Both these crabs reside in highly variable intertidal systems, typically in small pools with more potential for extreme variations in temperatures. They are also able to be out of the water for certain periods of time, either inadvertently or to find other, more suitable pools. These situations do not come without stress to the organism, however. The aim of this study is to determine the metabolic impact, as a proxy for stress, for hairy shore crabs and European green crabs at different water temperatures (14°C and 24°C) and after a period of desiccation. Metabolic rate of the crabs was measured using the change in fluorescence of during a 1-hour trial with the crabs submerged in resazurin, a dye which changes color as it oxidizes. Crabs were then placed under heat lamps for a 30-minute desiccation period, then placed back in resazurin for a follow-up 1-hour trial. We found that the temperature the crabs were held at before the desiccation period and the species had the greatest impact on the metabolic rate, with the European green crabs having a lower overall metabolic rate than the shore crabs, and the crabs held at 24°C being less sensitive to the desiccation period than the ones held at 14°C. This study can help to inform how each species of crab can cope with environmental stressors, and which crabs may be better adapted to handle future changes to the intertidal environment.

I. Introduction

European green crab (*Carcinus maenus*, further referred to as *Carcinus*) are known to be fierce competitors in intertidal systems, with an ability to adapt to a wide range of conditions, mainly salinity and temperature, which has made them rather adept to nearly any environment they find themselves in, the making for a highly invasive organism (Leignel et al 2014). Understanding how these crabs will interact with and potentially compete with native species, such as the hairy shore crab (*Hemigrapsus oregonensis*, further referred to as *Hemigrapsus*), has been a topic of rising interest as more *Carcinus* are found in the Salish Sea and as the conditions continue to be altered due to climate change. Future predictions have the overall sea surface temperature of the Salish Sea rising anywhere from 0.5-1°C by 2050, for a max average of 19.6°C and a minimum average of 8.7°C (Walker et al 2022), with the tidal pools *Carcinus* and *Hemigrapsus* having the potential to fluctuate more, due to their smaller volume and intermittent exposure to air.

There have been numerous studies on the effects of water temperature on both *Carcinus* and several *Hemigrapsus* species, including *oregonensis*, showing that these crabs do have a considerable capacity for acclimating to higher temperatures within the limits of the typical seasonal variation of their native and invaded ranges (Dehnel 1960 and Tepolt et al 2020). A handful of studies have been done on the effects of desiccation, finding metabolic rate of *Hemigrapsus sanguineus* can double when exposed to air for 1.5-hour trials (Griffen et al 2024), though few studies have focused on the combined effects of elevated temperature and desiccation stress, and fewer have done direct comparisons of *Hemigrapsus* and *Carcinus* metabolic rates when stressed. The aim of this study is to compare the metabolic rates of *Carcinus maenus* and *Hemigrapsus oregonensis* in response to both elevated temperature and desiccation to simulate what may occur in a real intertidal ecosystem. To do this, we will measure the metabolic rate of each crab using resazurin solution held at a control and elevated temperature, then expose the crabs to air, followed by a second trial in resazurin at the same two temperatures. These results will then be used as a proxy for the stress of the crabs, which will be compared.

II. Methods

To measure the amount of stress caused to the crabs during the trials, the metabolic rate as respiration rate, was used as a proxy. To accomplish this, we followed the procedure for using resazurin solution to measure the amount of oxygen being consumed from a respirometry chamber as described by Huffmeyer et al 2025. The *Hemigrapsus* were held in 6-well plates and the *Carcinus* in condiment cups with resazurin solution for 1-hour trials, with 200 μ L aliquots of resazurin being taken every 10 minutes and transferred to a 96-well plate, including a blank. The 6-well plates held 8 mL of resazurin, and the condiment cups held 40 mL; enough resazurin to cover the crabs completely, without there being so much that the change in color would not be observable. After the experiment was finished, the 96-well plates were read using a plate reader, and the results subsequent fluorescence values were normalized by subtracting the blank values and dividing the fluorescence at each time point by the weight of the crab. These values were graphed in R Studio, the slope of the line through the fluorescence time points being the metabolic rate. All statistical tests were also run in R Studio.

The overall goal was to simulate what may occur in an intertidal system where these two species live and often coexist. To do this, we held the resazurin dye at two different temperatures: 14°C and 24°C. 14°C is representative of the standard water temperature of the Puget Sound around mid-summer, and 24°C is considerably higher than water temperatures in the Puget Sound, though some tide pools can reach similar temperatures during summer months, depending on their location. Five *Hemigrapsus*, and one *Carcinus* were kept in each temperature, with this being repeated twice, for a total of 10 *Hemigrapsus* and two *Carcinus* in each temperature. After an hour in these temperatures, the crabs were taken out and placed in corrals with sandy substrate underneath heat lamps and partial shading to simulate a period of desiccation that may occur in the wild. The air temperature was held between 35-37.8°C, and the sand temperature held at 30°C. Crabs were held here for 30 minutes, then returned to resazurin at

their respective temperatures to measure the change in metabolic rate. Due to the limited number of *Carcinus* available, the data was bootstrapped with replacement in Excel to result in even numbers of observations between species.

III. Results

Treatment	Compared to	diff	lwr	upr	p adj	Significance
Hemigrapsus Post	Carcinus Post	44.9080	15.8308	73.9851	0.0112	*
Carcinus Pre	Carcinus Post	-0.0306	-29.1077	29.0456	1.0000	
Hemigrapsus Pre	Carcinus Post	40.2781	11.2009	69.3552	0.0166	*
Carcinus Pre	Hemigrapsus Post	-44.9385	-74.0157	-15.8614	0.0112	
Hemigrapsus Pre	Hemigrapsus Post	-4.6299	-33.7070	24.4473	0.9111	
Hemigrapsus Pre	Carcinus Pre	40.3086	11.2315	69.3858	0.0165	*
Hemigrapsus 14	Carcinus 14	35.4915	25.7636	45.2194	0.0004	***
Carcinus 24	Carcinus 14	0.0027	-9.7252	9.7306	1.0000	
Hemigrapsus 24	Carcinus 14	49.7278	39.9999	59.4556	0.0001	****
Carcinus 24	Hemigrapsus 14	-35.4888	-45.2167	-25.7609	0.0004	**
Hemigrapsus 24	Hemigrapsus 14	14.2363	4.5084	23.9641	0.0136	*
Hemigrapsus 24	Carcinus 24	49.7251	39.9972	59.4529	0.0001	****
24 C	14 C	-83.7558	-108.6320	-58.8796	0.0044	**
Hemigrapsus	Carcinus	2.4263	-253.2833	258.1348	0.9712	

Table 1. Results of the statistical analysis run on data collected. All tests conducted were two-way ANOVA with a follow-up Tukey test. * p-val ≤ 0.05 , ** p-val ≤ 0.005 , *** p-val ≤ 0.0005 , **** p-val ≤ 0.0001 .

From the experiment, we found significant differences in the metabolic rate between the *Hemigrapsus* and *Carcinus* at both temperatures, and at the different stages of the experiment (Pre- and Post-desiccation). The *Carcinus* had a consistently lower metabolic rate compared to the *Hemigrapsus*, which was not significantly between temperatures when compared to itself. However, there were no significant differences in metabolic rate for *Hemigrapsus* or *Carcinus* individually pre- and post-desiccation.

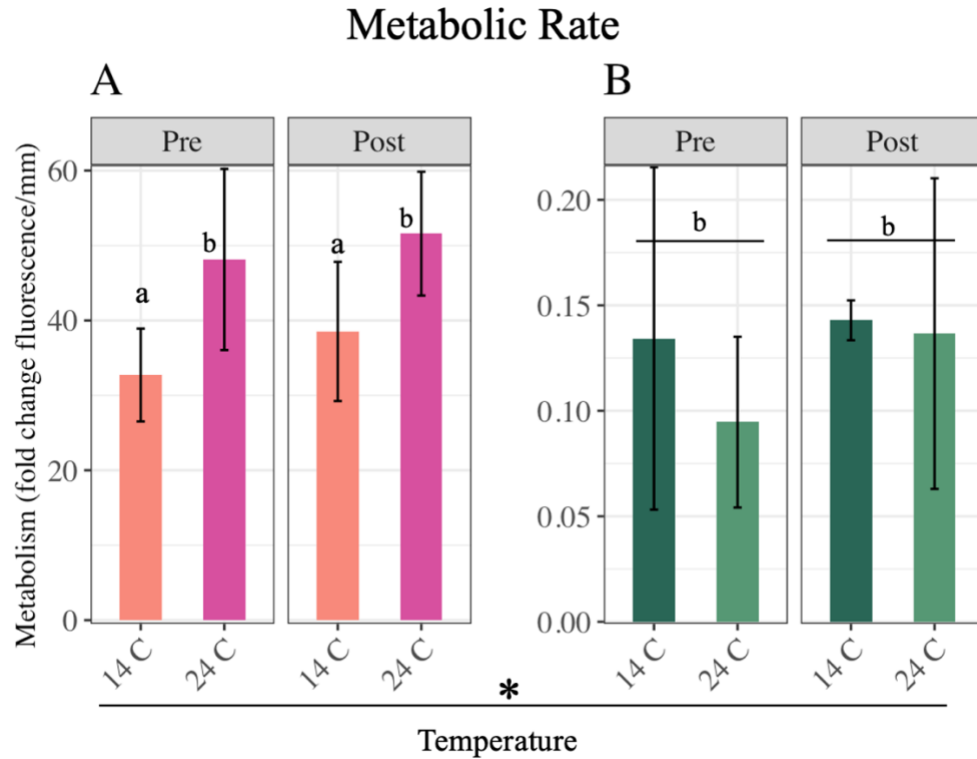


Figure 1: Mean metabolic rate as fold change in resazurin fluorescence/mm for *Hemigrapsus* (A) and *Carcinus* (B) at different temperatures before (Pre) and after (Post) desiccation. The y-axis scale for the *Carcinus* is lower than the y-axis scale for *Hemigrapsus* due to the much lower overall metabolic rate. “a” is significantly different from “b” (p-val < 0.05). * p-val < 0.05. Error bars indicate $\pm$ SD.

In addition, when comparing the changes in the metabolic rates before and after desiccation, we found no significant difference in the change depending on species, though there was a significant difference in the change depending on the temperature the crabs were held in, with the crabs held at 24°C having a smaller change in their metabolic rate compared to those held at 14°C.

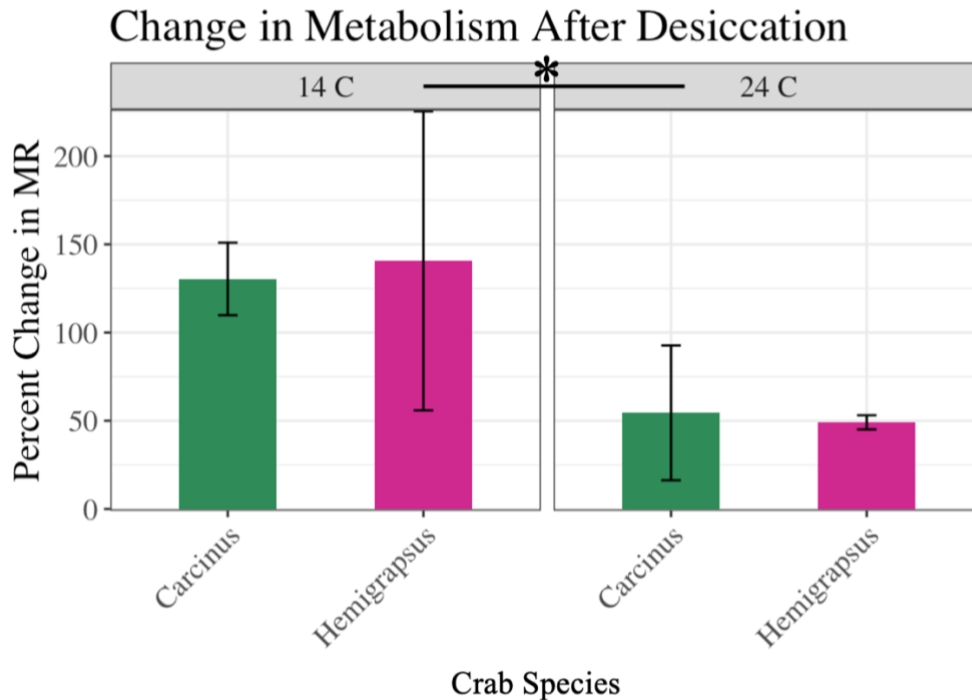


Figure 2. Percent change in metabolic rate after desiccation of each crab species for the different temperature trials. Error bars indicate $\pm$ SD. * p-val < 0.05 between the temperature trials.

IV. Discussion

Based on the results, we can conclude that the only factors that impacted the metabolic rate of the two species were the species themselves and the temperature they were held at, though only for *Hemigrapsus*. The change in metabolic rate after desiccation was much higher for the crabs held at a colder temperature compared to the warm treatment. This effect could have arisen from the fact the metabolic rate for the crabs was already elevated in the 24°C treatment, so the change was less pronounced when placed back in the warm resazurin. Crabs held at 14°C would have had to adjust their metabolic rate to cope with the hot air temperature in the desiccation period, resulting in the more dramatic change seen in Figure 2. With these dynamics in mind, we can interpolate how this can apply to actual intertidal ecosystems. Should either species of crab be in a cooler tidepool and experience desiccation, they could potentially be more vulnerable to stress caused by desiccation than those in warmer pools. In addition, the *Hemigrapsus* are more heavily impacted by temperature stress than the *Carcinus*, meaning they would potentially be less tolerant of future changes to the intertidal environment.

Future studies on this subject could focus on the effects of temperature and desiccation to a larger scale, such as having a longer desiccation period to better simulate what crabs may experience in a real intertidal environment. It could also be beneficial to measure multiple parameters that are used to measure stress, especially since resazurin could not be used during the desiccation period to directly measure metabolic rate.

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